

STAT230A Lecture 27 Interactions in OLS

1 Interactions in OLS

1.1 Interactions in OLS

考虑以下 OLS with interaction

$$y_i = \hat{\beta}_0 + \underbrace{\hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}}_{\text{main effects}} + \underbrace{\hat{\beta}_{12} x_{i1} x_{i2}}_{\text{interaction term}} + \hat{\varepsilon}_i$$

记作 $y \sim x_1 + x_2 + x_1 * x_2$ 或 $y \sim x_1 * x_2$

Logic ▾

首先考虑 simplest case: $x_{i1}, x_{i2} \in \{0, 1\}$

1.2 Theorem: Two Binary Covariates Interact

考虑 OLS with interaction:

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \hat{\beta}_{12} x_{i1} x_{i2} + \hat{\varepsilon}_i$$

其中 $x_{i1}, x_{i2} \in \{0, 1\}$

定义不同 group 的 $\bar{y}$ 为:

$$\begin{aligned} x_1 = 1, x_2 = 1 &\Rightarrow \bar{y}_{11} \\ x_1 = 1, x_2 = 0 &\Rightarrow \bar{y}_{10} \\ x_1 = 0, x_2 = 1 &\Rightarrow \bar{y}_{01} \\ x_1 = 0, x_2 = 0 &\Rightarrow \bar{y}_{00} \end{aligned}$$

则:

$$\begin{aligned} \hat{\beta}_0 &= \bar{y}_{00} \\ \hat{\beta}_1 &= \bar{y}_{10} - \bar{y}_{00} \\ \hat{\beta}_2 &= \bar{y}_{01} - \bar{y}_{00} \\ \hat{\beta}_{12} &= \bar{y}_{11} - \bar{y}_{10} - \bar{y}_{01} + \bar{y}_{00} \\ &= (\bar{y}_{11} - \bar{y}_{10}) - (\bar{y}_{01} - \bar{y}_{00}) \\ &= \underbrace{(\bar{y}_{11} - \bar{y}_{01})}_{\text{effect of } x_1 \text{ on } y \text{ when } x_2=1} - \underbrace{(\bar{y}_{10} - \bar{y}_{00})}_{\text{effect of } x_1 \text{ on } y \text{ when } x_2=0} \end{aligned}$$

Logic ▾

可以对上述结论进行推广, 考虑 a binary covariate interacts with a general covariate

1.3 A Binary Covariate Interacts with a General Covariate

Settings:

考虑 OLS with interaction:

$$y_i = \hat{\beta}_0 + \hat{\beta}_x x_i + \hat{\beta}_z z_i + \hat{\beta}_{xz} x_i z_i + \hat{\varepsilon}_i$$

其中:

- y_i 为 continuous
- x_i 为 continuous
- z_i 为 binary

Model Fitting:

此时有以下两个 regressions:

$$\hat{y}_i = \begin{cases} \hat{\beta}_0 + \hat{\beta}_x x_i & \text{if } z_i = 0 \\ \hat{\beta}_0 + \hat{\beta}_z + (\hat{\beta}_x + \hat{\beta}_{xz})x_i & \text{if } z_i = 1 \end{cases}$$

因此上述 OLS with interaction 等价于 fit 以下 regressions:

- Fit $y \sim x$ in $z = 0$ group
- Fit $y \sim x$ in $z = 1$ group

Interpretation:

在 NLM / HLM 下, z 对于 y 的 effect 可以理解为以下 heterogeneous treatment effects:

$$\mathbb{E}[y_i | z_i = 1, x_i] - \mathbb{E}[y_i | z_i = 0, x_i] = \beta_z + \beta_{xz}x_i$$

其对不同的 x_i 取不同的值

Report:

对于 model without interaction, 我们可以考虑 report

$$\frac{\partial \mathbb{E}[y_i | z_i, x_i]}{\partial z_i} = \beta_z$$

类似地, 对于可以 model with interaction, 有

$$\frac{\partial \mathbb{E}[y_i | z_i, x_i]}{\partial z_i} = \beta_z + \beta_{xz}x_i$$

因此我们考虑 report average estimated derivatives across individuals, 即

$$\text{APE} := \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_z + \hat{\beta}_{xz}x_i) = \hat{\beta}_z + \hat{\beta}_{xz}\bar{x}$$

其也被称为 **average partial effect (APE)** 或 average marginal effect

⚠ Remark: Centering 对 Model with Interaction 的影响 ∨

- 若我们在 fit regression 之前对 x 进行 centering, 即 $x_i \rightarrow x_i - \bar{x}$, 则此时 $\hat{\beta}_2 = \text{APE}$
- 若我们在 fit regression 之前对 x 和 z 进行 centering, 则此刻 main effect = APE

这与我们之前在 FWL theorem 处的描述 (centering covariates 不影响 coefficient values, 只影响 intercept) 不同, 对于 model with interaction, centering x 会产生以下影响:

	Intercept	Main Effect	Partial Effect
Model with Interactions	disappear	unchanged	
Model without Interactions	disappear	change	unchanged

由此可见, 加入 interaction 会使 interpretation 变得混乱

🔗 Logic ∨

加入 interaction 除了会使 interpretation 变得混乱, 还有其他缺点

1.4 Interaction 的缺点

Power:

相较于 main effect, 难以通过 hypothesis testing 对 interaction 进行 detection

Example: Two Binary Covariates Interact ▾

考虑 two binary covariates interact 的情况:

$$\hat{\beta}_{12} = (\bar{y}_{11} - \bar{y}_{10}) - (\bar{y}_{01} - \bar{y}_{00})$$

此时有

$$\text{Var}[\hat{\beta}_{12}] = \frac{\sigma_{11}^2}{n_{11}} + \frac{\sigma_{10}^2}{n_{10}} + \frac{\sigma_{01}^2}{n_{01}} + \frac{\sigma_{00}^2}{n_{00}}$$

因此即便 $n = n_{11} + n_{10} + n_{01} + n_{00}$ 很大, 只要其中任意一项很小, $\text{Var}[\hat{\beta}_{12}]$ 就会很大

Model Selection:

Interactions 和 model selection, Lasso, Ridge 并不兼容

Example ▾

若我们考虑 X matrix with all possible interactions, 并进行 model selection, 此时即便 $x_1 * x_2$ 和 $x_1 * x_3$ 被纳入 model, 也不能保证 x_1 被纳入 model

Remark: ▾

一种潜在的解决方法是先进进行 model selection, 再对选出的 covariants 添加 interactions